

Toward a Rigorous Path to Solving the Hodge Conjecture - Nickolas Kusman

The **Hodge Conjecture** is a central open problem in algebraic geometry that asserts: *On any smooth complex projective variety, every rational cohomology class of pure Hodge type (p,p) is a **rational** linear combination of classes of algebraic cycles of codimension p* [jmilne.org](https://en.wikipedia.org/wiki/Hodge_conjecture). In other words, the topological “Hodge” pieces of the cohomology (the so-called *Hodge cycles*) should actually come from genuine algebraic subvarieties. This conjecture connects the continuous world of **Hodge theory** (decomposition of cohomology) with the discrete world of **algebraic cycles** (subvarieties), and it has been designated one of the Clay Millennium Prize Problems[claymath.org](https://www.claymath.org/). Below we explore several frameworks that have guided progress on the Hodge Conjecture, outlining how each provides either partial results (reducing the general conjecture to tractable cases) or new structural constraints that a eventual proof might exploit.

Classical Approaches: Hodge Decomposition and Lefschetz Theorems

Hodge’s original ideas and Lefschetz’s results. The conjecture arose from Hodge’s 1941 discovery that on a complex projective variety X , the singular cohomology splits as $H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X)$ (Hodge decomposition). In particular, any complex subvariety of complex codimension p yields a cohomology class in $H^{p,p}(X)$, and Hodge observed that the *Lefschetz $(1,1)$ -theorem* already confirmed the conjecture for $p=1$ [en.wikipedia.org](https://en.wikipedia.org/wiki/Lefschetz_(1,1)-theorem). Lefschetz’s theorem (proved in 1924) says that any cohomology class of type $(1,1)$ on a projective variety is the first Chern class of some holomorphic line bundle, i.e. it is the class of an algebraic divisor[en.wikipedia.org](https://en.wikipedia.org/wiki/Lefschetz_(1,1)-theorem). This settles the Hodge Conjecture for degree 2 cohomology ($p=1$ case) completely. More generally, using the Hard Lefschetz theorem (which implies a powerful symmetry on the Hodge structure), one can reduce other cases to the $(1,1)$ case: if the conjecture holds for classes of degree $2p$, it also holds for classes of complementary degree $2(\dim X - 2p)$ [en.wikipedia.org](https://en.wikipedia.org/wiki/Hodge_conjecture). Combining these facts, it is known that *the Hodge Conjecture holds for all varieties of complex dimension ≤ 3* [en.wikipedia.org](https://en.wikipedia.org/wiki/Hodge_conjecture) (since any Hodge class on a threefold is either in degree 2 or 4, both covered by the above reductions). These results exemplify how classical tools reduce the scope of unknown cases – essentially only the “middle-dimension” cohomology in higher-dimensional varieties remains mysterious[en.wikipedia.org](https://en.wikipedia.org/wiki/Hodge_conjecture).

Normal functions and limitations of the classical inductive approach. A natural classical strategy is **induction on $\dim X$** by slicing X with hyperplanes. If $H \subset X$ is a hyperplane section (also a projective variety of dimension $\dim X - 1$), there are maps relating cohomology and cycles on H to those on X (Lefschetz hyperplane theorem, etc.). In the 1970s, Zucker, Griffiths and others developed the theory of **normal functions** (sections of the

intermediate Jacobian bundle) to try to lift algebraic cycles from a hyperplane section H to a cycle on X . In principle, if one assumes the Hodge Conjecture is true for the hyperplane H , one can try to construct a cycle on X representing a given Hodge class by deforming a cycle on H . However, a serious obstruction was discovered: one needs not only cycles on H , but also that *all* classes in the **intermediate Jacobian** $J^p(H)$ (which measures the difference between Hodge classes and algebraic cycles via Abel–Jacobi theory) come from algebraic cycles on H claymath.org. In many cases this fails – there can be “transcendental” elements in the intermediate Jacobian of H (e.g. infinitude of certain cycle invariants claymath.org), ruining the simple inductive step. In fact, it is known that the group of algebraically trivial cycles on some varieties can have infinite rank, which means the classical method of iteratively slicing by hyperplanes cannot straightforwardly prove the conjecture in general claymath.org. This underscores the need for deeper insights beyond the reach of classical Lefschetz-type theorems.

Topological and smooth variants. It is worth noting that the Hodge Conjecture is delicate in requiring **algebraic** cycles. If one allows **analytic** subvarieties or singular currents, the problem becomes easier (by Hodge theory any (p,p) cohomology class can be represented by a closed (p,p) -current). But Chow’s theorem ensures that in projective varieties “analytic subvariety” actually means an algebraic subvariety, so the distinction is in allowing limits of cycles. A famous warning example by Atiyah–Hirzebruch showed that the *integral* version of the Hodge conjecture is false: there exist torsion cohomology classes of type (p,p) that provably do not come from any algebraic cycle. These rely on smooth but highly non-algebraic constructions (like *exotic* manifolds). Thus one sticks to **rational coefficients** in the conjecture. Mumford also exhibited cases of compact Kähler varieties (not projective) with Hodge classes that are not algebraic. All these examples reinforce that the projectivity assumption is crucial and that any successful path must use algebraicity in an essential way.

Summary of classical outcomes: In short, classical Hodge theory provides the *foundation* of the conjecture (the Hodge decomposition) and proves it in low codimensions and low dimensions en.wikipedia.org. Tools like the Lefschetz theorem on $(1,1)$ -classes and the Hard Lefschetz theorem reduce many cases to the hard “middle-dimensional” case. However, purely classical approaches hit a wall due to intermediate Jacobian obstructions claymath.org. The failure of an easy inductive proof led mathematicians to enrich the toolkit over the past decades, as we discuss next.

Modern Hodge-Theoretic Developments: Mixed Hodge Structures and Motives

Mixed Hodge structures and extensions. In the 1970s, Deligne introduced the theory of **mixed Hodge structures (MHS)** which generalizes Hodge decomposition to *non-pure* situations (e.g. singular or open varieties, or relative cohomology). A mixed Hodge structure has a weight filtration $W_{\bullet}$ and Hodge filtration $F^{\bullet}$ that together encode how a cohomology class can arise as a limit or extension of pure Hodge structures. The relevance to

the Hodge Conjecture is that a rational Hodge class (a class $\gamma \in H^{2p}(X, \mathbb{Q})$ with $[\gamma] \in H^{p,p}(X)$) often gives rise to a nontrivial **extension** of Hodge structures. Specifically, given a Hodge class γ , one can consider an algebraic cycle Z approximating it and form a **normal function** (Abel–Jacobi invariant) measuring the difference. This invariant lives in an intermediate Jacobian $J^p(X)$, which can be identified with an extension group of Hodge structures: $J^p(X) \cong \text{Ext}^1_{\text{MHS}}(\mathbb{Z}(0), H^{2p-1}(X, \mathbb{Z}(p)))$ (the extension of a trivial Hodge structure by a pure one) claymath.org. The *Hodge Conjecture for class γ* is true if and only if this extension splits, i.e. the normal function vanishes. In practice, this means modern Hodge theory provides **criteria** to detect when a Hodge class is algebraic: it must lie in the kernel of all Abel–Jacobi maps. Using MHS tools, one can sometimes verify this for special classes. More generally, mixed Hodge theory constrains how *families* of Hodge classes behave:

Cattani–Deligne–Kaplan’s theorem (1987) shows that the locus in a family of varieties where a given rational class remains of type (p,p) is an algebraic (in fact, arithmetic) subvariety of the base claymath.org. This algebraicity of the so-called *Hodge loci* is a nontrivial fact purely from Hodge theory – it does not assume the Hodge Conjecture, but it is **consistent with it**. It implies that the condition “ γ is a Hodge class” is as restrictive as an algebraic equation in moduli, hinting that “ γ comes from a cycle” might also be such a condition. Recent advances have strengthened this: using o-minimal techniques, Bakker, Klingler, Tsimerman and others (2018–2020) proved that not only are Hodge loci algebraic, but their *closures* in moduli are also algebraic (fulfilling a conjecture of Griffiths). These results don’t prove any new cases of the Hodge Conjecture, but they *reduce the possibilities* for counterexamples by showing Hodge classes cannot appear “randomly” in families – they obey arithmetic constraints.

Saito’s Mixed Hodge Modules – functorial control of Hodge theory. A landmark development was Morihiko Saito’s theory of **Mixed Hodge Modules (MHM)** in the 1980s. This is a highly technical framework that packages Hodge structures together with **perverse sheaves** (from algebraic topology) and **$\mathbb{D}$ -modules** (from algebraic analysis) into a six-functor formalism paralleling ℓ -adic étale theory en.wikipedia.org. In essence, Saito managed to attach a well-behaved mixed Hodge structure to *any* derived object such as the intersection cohomology of a singular variety en.wikipedia.org. For the Hodge Conjecture, this means one can now cleanly study the cohomology of singular hyperplane sections or degenerations with Hodge-theoretic tools. For example, the failure of the naive induction via hyperplanes might be tackled with MHM by allowing *singular* hyperplane sections and keeping track of the Hodge structure of their intersection cohomology. The **Decomposition Theorem** (Beilinson–Bernstein–Deligne 1982, with M. Saito adding Hodge structure compatibly) ensures that the pushforward of the constant Hodge module on X splits into a direct sum of pure Hodge modules on the base. Concretely, if $f: X \rightarrow S$ is a morphism (like a Lefschetz pencil), one gets a **filtration (perverse filtration)** on $H^*(X)$ whose graded pieces are precisely the intersection cohomology of the strata of singular fibers, each of which carries a pure Hodge structure by Saito’s theory. This structure is so rich that it severely limits how a (p,p) class on X can behave: as one moves in the pencil, any would-be Hodge class must either come from the base or specialize from a singular fiber class. Although this has not yet led to a full proof, it **reduces the conjecture to showing certain pieces of these Hodge modules correspond to**

actual cycles. The power of MHM is that it is **compatible with all standard operations (pullback, pushforward, duality)**, so one can attempt sophisticated inductive or specialization arguments that were impossible to control with classical methods. In summary, Saito's framework is a tool to "track" Hodge classes through geometric transitions.

Motives, periods, and Deninger's insights. Grothendieck's ambitious "Theory of Motives" is closely related to the Hodge Conjecture. A *motive* can be thought of as a "universal" realization of a variety's cohomology, encoding all cohomology theories (singular, ℓ -adic, de Rham, etc.) simultaneously claymath.org. Grothendieck formulated the *Standard Conjectures* on algebraic cycles (such as the Lefschetz standard conjecture) that are even stronger than the Hodge Conjecture and would imply it. One standard conjecture posits that certain canonical cohomology idempotents (the Künneth projectors) are algebraic; if all those were known, one could argue that any Hodge class is basically a piece of the diagonal class and hence algebraic claymath.org. In fact, Deligne pointed out that assuming those standard conjectures, the natural functor from the category of pure motives to that of pure Hodge structures becomes fully faithful claymath.org. In simple terms: *if all the "obvious" cycles exist, then no further Hodge cycles can exist beyond those generated by algebraic ones*, so the Hodge Conjecture holds automatically. This reduction is conceptually important – it shows the conjecture is a part of a much bigger package of conjectures. While the Standard Conjectures remain open in general, they have been proven in certain cases (e.g. for surfaces, abelian varieties of CM type, some K3 surfaces via work of André, Tankeev, etc.), thereby establishing the Hodge Conjecture for those cases as well.

Another modern viewpoint comes from the study of **period integrals**. A *period* is an integral of an algebraic differential form over a cycle in homology; Hodge theory tells us these periods encode the Hodge decomposition. Deninger and others have observed that if a Hodge class is algebraic, then certain period integrals must take special values (often related to zeta or L -function values) annals.math.princeton.edu. For example, the Hodge Conjecture implies that the periods of any variety with a complex multiplication (CM) motive are products of Gamma-values (generalizing the classical Chowla–Selberg phenomenon for CM elliptic curves) annals.math.princeton.edu. **Conversely, if one could show such a period relation holds, it would strongly indicate the existence of the corresponding cycle.** This is a two-way street: assuming the Hodge Conjecture, one can *predict* and then attempt to prove deep formulas for period integrals; or conversely, leverage known cases of such formulas as evidence for the conjecture. A concrete success of this interface is the proof of special cases of the **Gross–Deligne period conjecture**: Maillot and Rössler (2004) proved Gross–Deligne for a class of motives by showing that the necessary period identities hold in the presence of an automorphism of finite order annals.math.princeton.edu. Their proof used techniques of Arakelov geometry and was partly motivated by the expectations from Hodge theory. While this does not prove any new case of Hodge Conjecture outright, it illustrates that *Hodge classes impose extra arithmetic conditions*. In short, **modern developments have tied the Hodge Conjecture to the web of "motivic" conjectures in number theory**: absolute Hodge cycles (those invariant under all isogenies and Galois actions) are expected to equal algebraic cycles jmilne.org, the conjecture holds for all abelian varieties with CM (by reduction to the Tate conjecture via

Honda–Tate theory), and failure of the Hodge Conjecture would likely force surprising failures in these other arenas.

Derived Categories, Perverse Sheaves, and Motivic Cohomology

From cohomology to derived categories. Instead of working directly with cohomology groups, one can consider the **derived category of coherent sheaves** $D^b(X)$ or related derived categories as an invariant of X . These categories capture subtle geometric information (through objects like structure sheaves of subvarieties, complexes, etc.) not visible in just the cohomology ring. One intriguing direction is that certain **equivalences or semiorthogonal decompositions** of derived categories can imply relationships between cycles. For instance, Kuznetsov showed that a cubic fourfold $X \subset \mathbb{P}^5$ has an associated K3 surface category; when this category actually comes from an *actual* K3 surface, X acquires an algebraic class in $H^{2,2}(X)$ (coming from the K3 surface’s algebraic cycles) which is conjectured to be equivalent to X being “special” (possibly rational). This kind of **derived-equivalence criterion** indicates that sometimes the existence of a cycle is equivalent to a categorical condition. In general, the *presence of an additional Hodge class often signals an “extra” structure on the variety*, which could be a derived category splitting. While these observations are case-specific, they hint that derived categories (which are more flexible than Chow groups) might be a vehicle to *construct* cycles. If one can identify an object in $D^b(X)$ whose Chern classes produce a given Hodge class, that object would serve as an “algebraic witness” to the class. Recent work on **non-commutative Hodge conjectures** (seeking analogues of Hodge classes in derived categories of noncommutative motives) also suggests that t -structures on triangulated categories of motives might be chosen so that their “heart” yields the right group of cycles degruyter.com/projecteuclid.org.

Perverse sheaves and filtrations by support (coniveau). The conjecture can be reformulated as a statement about the **coniveau filtration** on cohomology: $N^p H^{2p}(X, \mathbb{Q})$ consists of classes vanishing on a subvariety of codimension p , and the Hodge Conjecture asserts that $H^{p,p}(X) \cap H^{2p}(X, \mathbb{Q}) = N^p H^{2p}(X, \mathbb{Q})$ for projective X . This is a statement about how cohomology “comes from subvarieties”. **Perverse sheaves** (from the decomposition theorem mentioned earlier) naturally yield a filtration on cohomology by the dimension of support of cycles. In fact, the *perverse Leray filtration* associated with a map $f: X \rightarrow Y$ is closely related to the classical coniveau filtration. One approach is to use a **perverse t -structure on the category of motives** (still conjectural to construct rigorously) so that the **graded pieces of this t -structure correspond to subvariety-supported cohomology**. If such a theory were in place, proving the Hodge Conjecture might reduce to checking that certain Ext groups vanish in the abelian category of motives. There is active research in this direction: *Nori’s constructible abelian category of mixed motives* and more recent advances by Wildeshaus, Déglise, and others aim to realize a motivic t -structure whose heart would be something like “perverse motivic sheaves” cambridge.org/math.columbia.edu. The **motivic cohomology** groups (à la Voevodsky,

or equivalently Bloch's higher Chow groups) provide a graded-object that maps to usual cohomology via a **motivic spectral sequence** math.columbia.edu. On the E_2 -page of this spectral sequence lie the Chow groups (cycles) and their higher analogues, and it converges to singular cohomology. The Hodge Conjecture is essentially the claim that for *pure weight* cohomology, this spectral sequence collapses at E_2 for the (p,p) part – no further differentials are needed to obtain those classes. In practical terms, one needs to show that any cohomology class with Hodge numbers (p,p) can be lifted to an *actual cycle* in the E_2 stage. If some differential d_r in the spectral sequence killed that class (meaning it was “transcendental” and only appeared at later stages), then the Hodge Conjecture fails. Researchers study specific cases of this sequence: for example, Bloch and Srinivas showed that if a certain “bloated” Jacobian (an Ext group in motives) vanishes, then the conjecture holds for that variety. Beilinson formulated conjectures predicting that these motivic Ext groups are governed by values of L -functions, and thus if the L -functions behave (e.g. have the expected zeros), the necessary Ext might vanish, yielding the needed cycle. **In summary, derived-category and motivic methods tackle the Hodge Conjecture by reframing it as a statement about the structure of an entire category or spectral sequence**, rather than individual cycles. They introduce powerful algebraic and homological tools (like t -structures, Ext groups, and spectral sequences) that either simplify special cases or at least **constrain how a counterexample would have to look** (often in ways that seem impossible under other conjectures).

As a concrete example of a reframing, R. Thomas (2003) gave an equivalence: *The Hodge Conjecture is true for all varieties if and only if for every even-dimensional projective variety X^{2n} and every class $A \in H^{n,n}(X)$, one can find a sufficiently high-degree **nodal hyperplane section** $D \subset X$ such that the restriction $A|_D$ comes from a homology class of D* arxiv.org. Intuitively, instead of finding an algebraic cycle for A directly in X , this asks to find a divisor D with only ordinary double-point singularities where A “lives in the homology of D ”. If true, then by induction one eventually finds an actual cycle. This equivalence turned the question into a more geometric existence statement about divisors with many nodes. Such reformulations are valuable: even if they are hard to prove, they often reveal new necessary conditions. In Thomas's case, it suggests studying **singular hypersurfaces** and their vanishing cycles as a route to producing algebraic cycles – a strategy that intersects with symplectic geometry and singularity theory (via techniques of Lagrangian vanishing cycles and Picard–Lefschetz theory). Indeed, progress on bounding the number of nodes or understanding the distribution of vanishing cycles could feed back into Hodge theory.

Post-2015 Algorithmic and New Frameworks

In recent years, there has been a growing interest in **algorithmic, computational, and “experimental” approaches** to problems like the Hodge Conjecture, as well as entirely new frameworks from analysis and tropical geometry that offer fresh perspectives:

- **Explicit cycle construction and verification.** With modern computer algebra systems, one can attempt to **find algebraic cycles** on specific varieties by solving polynomial

equations. For example, for certain hypersurfaces or modular varieties, algorithms can search for low-degree subvarieties. While this does not prove the conjecture in general, it can provide evidence in hard cases. A famous case is the **cubic fourfold** (a hypersurface in $\mathbb{P}^5$) where one expects that for special equations defining X , an extra algebraic surface exists inside X . These have been confirmed in many cases by finding those surfaces explicitly or by counting points over finite fields to detect their presence. Indeed, via the **Tate Conjecture** (the analog of Hodge Conjecture in ℓ -adic cohomology for varieties over finite fields), one can often verify algebraicity of a given Hodge class by checking it is fixed by Frobenius on a good reduction of X . *If an alleged counterexample to Hodge existed, one could attempt to find a prime of good reduction and test the Tate conjecture there – no such counterexample has been detected despite extensive computational searches.* For K3 surfaces and certain Calabi–Yau threefolds, point-counting algorithms have been used to successfully determine the rank of the Picard group (algebraic cycles of codimension 1) and have always aligned with Hodge-theoretic predictions. This computational aspect, while not a proof, serves to **reduce the conjecture to finite checks for specific families**.

- **Demailly’s analytic approach and positive currents.** Jean-Pierre Demailly in the 2010s formulated an approach using **transcendental analysis**: he showed that the Hodge Conjecture is *equivalent* to a statement about approximating certain closed currents by algebraic cycles [semanticscholar.org](https://www.semanticscholar.org/). Specifically, he proved that if every closed positive (p,p) -current with rational cohomology class could be approached by currents of integration on algebraic cycles, then any rational Hodge class would be algebraic. This turns the problem into solving a kind of PDE or extremal problem: one wants to “bend” a closed differential form into an actual algebraic cycle. Demailly even proposed a continuity method (a complex Monge–Ampère type equation) to construct such approximations typeset.io. Although a full implementation of this plan remains elusive, it introduced new **analytic constraints**. For instance, it is known that any would-be counterexample variety must carry a certain type of extremal positive current not approximable by algebraic ones; but extremal positive currents are quite special (e.g. they are *laminar* or foliate by subvarieties in some cases). Studying these analytic objects might eventually corner the conjecture by showing such currents cannot exist except when there is an algebraic cycle.
- **Tropical and combinatorial geometry.** Tropical geometry provides polyhedral analogues of algebraic varieties. In a 2019 work, Babaee and Huh used **tropical geometry** to investigate a “strongly positive” version of the Hodge conjecture. They introduced a tropical Hodge **Laplacian** and showed that certain natural guesses for a generalized Hodge statement fail in that context scholarworks.calstate.edu. In particular, they disproved a overly optimistic generalized Hodge conjecture for positive currents by constructing counterexamples via tropicalization scholarworks.calstate.edu. While a negative result, it is extremely valuable: it tells us which strengthened forms of Hodge Conjecture are *too strong* to be true, so that any proof must be more subtle. On the constructive side, tropical techniques might help to explicitly build cycles in degenerate

situations; a tropical cycle can sometimes be “algebraized” to a genuine cycle. The combinatorial constraints from tropical geometry (such as balancing conditions, discrete analogues of the Hodge decomposition, etc.) add yet another layer of necessary conditions for Hodge classes to be algebraic.

- **New reformulations and constraints.** The quest for a proof has generated many reformulations. Some, like Thomas’s nodal divisor criterion mentioned above, give a more accessible target (find a nodal hypersurface through a given cycle). Others link to different fields: for example, one could use **symplectic geometry** – Donaldson proved that any projective variety has lots of approximately holomorphic divisors (symplectic hypersurfaces) filling its homology. If one could tune Donaldson’s construction to land in a specific homology class, one might get the desired algebraic cycle or at least a homology representative very close to it. This is at heart an **algorithmic procedure** (solving increasingly large systems of equations to approximate a given class). Each such idea provides a **constraint**: any obstruction to these auxiliary problems (finding nodal divisors, solving a differential equation, etc.) would also obstruct the Hodge Conjecture. So far, no insurmountable obstruction has been found — which is consistent with the conjecture being true.

In conclusion, **progress on the Hodge Conjecture has come from a broad spectrum of techniques**, each providing pieces of a vast puzzle. Classical Hodge theory and algebraic geometry give us a strong base of known cases and reduction steps en.wikipedia.org. Modern Hodge module theory and the theory of motives embed the problem in rich structural frameworks that link it to deep arithmetic conjectures and often reduce the conjecture to more tractable statements about **filtrations, Ext groups, or functors** claymath.org. Derived categories and perverse sheaves suggest that what looks like a single cohomology class problem might be solved by understanding an entire **category of geometric or analytic objects**. Finally, new-age approaches – analytic, computational, tropical – both test the conjecture in extreme regimes and provide fresh intuition (or cautionary counterexamples to naive extensions scholarworks.calstate.edu). A successful solution to the Hodge Conjecture will likely synthesize insights from many of these approaches. For example, one could imagine an inductive proof that uses Saito’s mixed Hodge module machinery to reduce to showing the existence of a cycle in a limiting singular variety, then uses an analytic argument à la Demailly to produce the cycle in that limit, with the final input being a motivic vanishing conjecture or computation verified by computer for a specific small case. While this speculative path combines many deep theories, it is precisely the **interconnection** of these frameworks that has steadily pushed the boundary of what we understand about Hodge classes. Every partial result or new equivalence further “boxes in” the potential counterexamples, making a compelling case that the Hodge Conjecture — subtle as it is — should indeed be *true*. Each tool sharpens the conjecture, bringing us closer to the moment when a final piece unlocks the proof and we can declare every Hodge (p,p) class to be algebraic, as Hodge originally envisaged.

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